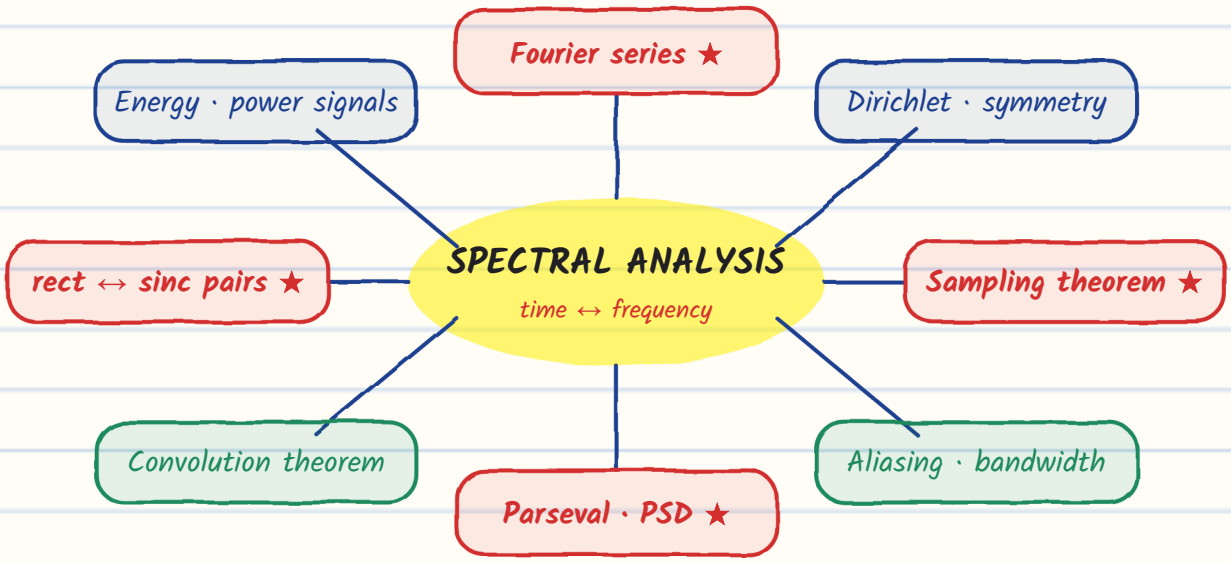
Spectral Analysis

Fourier series · Fourier transform · Convolution · Parseval · Sampling: exam-ready in 12 sheets



↑ the whole topic on one page: red = high-yield, green = method / applied

➜ Syllabus coverage

"Spectral analysis: Fourier series, sampling function, power spectrum, Fourier transform, convolution, Parseval's theorem". These notes add signal classification, the sampling theorem with aliasing, and bandwidth. Entropy and channel capacity are in the Information Theory lesson; AM/FM, PCM and TDM/FDM are in the Modulation & Multiplexing lesson.

High-yield list

① Nyquist rate = $2f_{max}$
② rect ↔ sinc, δ ↔ 1
③ even → cosines only
④ convolution ↔ product
⑤ Parseval: E in t = E in f
⑥ alias $f_a = |f - k \cdot f_s|$

Legend yellow = key term · red circle = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

➜ Contents

✓ Signals: classes, energy, power ..... p. 2
✓ Periodicity · Fourier series forms ..... p. 3
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1 Signals & Their Classification MCQ + Written

Definition A **signal** $x(t)$ is a function of time that carries information. **Spectral analysis** describes it by its **frequency content** (spectrum) instead of its waveform; Fourier analysis is the bridge.

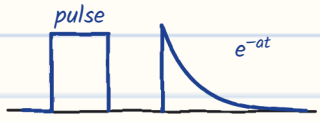
Pair	Test	Example
Continuous / discrete-time	every t / only $t = nT_s$	speech $x(t)$ / samples $x[n]$
Analog / digital	any amplitude / finite levels	mic output / PCM bits
Deterministic / random	formula / statistics only	$A\cos\omega_0 t$ / noise
Periodic / aperiodic	$x(t + T_0) = x(t)$ for all t	sine wave / one pulse
Even / odd	$x(-t) = x(t)$ / $x(-t) = -x(t)$	$\cos\omega_0 t$ / $\sin\omega_0 t$

2 Energy & Power Signals high-yield

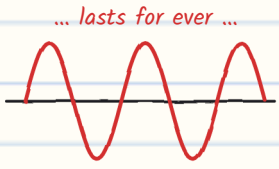
Formulas (1 Ω) $E = \int |x(t)|^2\,dt$ (J) · $P = \lim_{T\to\infty} (1/T) \int_{-T/2}^{T/2} |x(t)|^2\,dt$ (W)

energy signal: $0 < E < \infty$, $P = 0$ · **power signal**: $0 < P < \infty$, $E = \infty$

Signal	E	P
$A\,\mathrm{rect}(t/\tau)$	$A^2\tau$	0
$e^{-at}u(t)$, $a > 0$	$1/(2a)$	0
$A\cos(\omega_0 t + \theta)$	∞	$A^2/2$
DC level A	∞	A^2
unit step $u(t)$	∞	1/2
$t\cdot u(t)$, $e^{at}u(t)$	∞	∞ (neither)



energy: E finite, $P = 0$



power: P finite, $E = \infty$

Exam trap Every periodic signal is a **power** signal, never an energy signal. A signal cannot be both; some are **neither** ($t\cdot u(t)$). Phase θ and frequency do not change the power of $A\cos(\omega_0 t + \theta)$.

➜ Periodicity, even/odd parts & the power of a sum

Periodicity of a sum: $x_1 + x_2$ is periodic only if T_1/T_2 is rational; then $T_0 = LCM(T_1, T_2)$. $\cos(2\pi t/3) + \sin(2\pi t/5) \rightarrow T_0 = 15$ s; $\cos t + \cos \pi t \rightarrow$ ratio $\pi \rightarrow$ **not periodic.**

Even/odd parts: $x_e = [x(t) + x(-t)]/2$, $x_o = [x(t) - x(-t)]/2$. even × even = even, odd × odd = even, even × odd = odd; $\int_{-a}^{a}$ (odd) $dt = 0$.

Memory trick "**Energy signals die out, power signals live for ever.**" Finite duration or decaying → energy; repeating or constant → power.

Worked example ✍ $x(t) = 3\cos(100\pi t) + 4\sin(200\pi t)$: $f = 50$ Hz and 100 Hz, so $T_0 = 20$ ms. Powers add (different frequencies): $P = 9/2 + 16/2 =$ **12.5 W**

3 Fourier Series: Three Equivalent Forms

high-yield

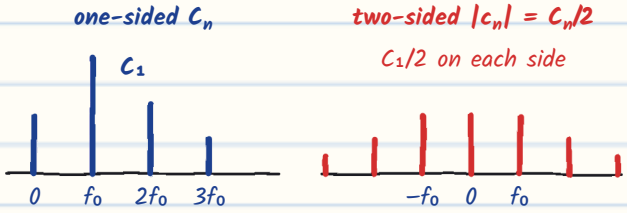
Definition A periodic signal $x(t)$ with period T_0 (meeting Dirichlet's conditions) equals a **DC term plus harmonics** at nf_0, where $f_0 = 1/T_0$ and $\omega_0 = 2\pi/T_0$. Its spectrum is a set of discrete lines (a line spectrum).

Form	Series	Coefficients
Trigonometric	$a_0 + \sum_{n=1}^{\infty} [a_n \cos n\omega_0 t + b_n \sin n\omega_0 t]$	a_0, a_n, b_n (box below)
Compact (polar)	$C_0 + \sum C_n \cos(n\omega_0 t + \theta_n)$	$C_n = \sqrt{(a_n^2 + b_n^2)}$, $\theta_n = -\tan^{-1}(b_n/a_n)$
Exponential	$\sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$	$c_n = (1/T_0)\int_{T0} x(t)\, e^{-jn\omega_0 t}\, dt$

Coefficients (∫ over one period) $a_0 = (1/T_0)\int x(t)\, dt$ = DC (average) · $a_n = (2/T_0)\int x(t) \cos n\omega_0 t\, dt$ · $b_n = (2/T_0)\int x(t) \sin n\omega_0 t\, dt$

Links: $c_0 = a_0$ · $c_n = (a_n - jb_n)/2$ · real $x(t)$: $c_{-n} = c_n^* \Rightarrow$ **$|c_n| = C_n/2$** $(n \geq 1)$

Exam trap In the **two-sided** (exponential) spectrum each line at $\pm nf_0$ is **half** the one-sided amplitude C_n; only the DC line keeps its height. Negative frequencies are a mathematical bookkeeping device, not a physical band.



4 Dirichlet Conditions & Symmetry

MCQ favourite

Dirichlet conditions (sufficient, not necessary), per period:

1. absolutely integrable: $\int_{T0} |x(t)|\,dt < \infty$
2. finite number of maxima and minima
3. finite number of **finite** discontinuities

At a jump the series converges to the **mid-point** $[x(t^+) + x(t^-)]/2$.

Symmetry	What survives
Even: $x(-t) = x(t)$	a_0, a_n; $b_n = 0$
Odd: $x(-t) = -x(t)$	b_n only
Half-wave: $x(t \pm T_0/2) = -x(t)$	**odd n only**
Even + half-wave	odd cosines only

Memory trick **Even ↔ cos, Odd ↔ sin** (cos is even, sin is odd). "**Half**-wave symmetry keeps the **odd** harmonics."

Shortcuts Adding a DC level changes only a_0. A time shift changes only the **phases** θ_n, never $|c_n|$. Scaling time changes f_0, not the coefficients.

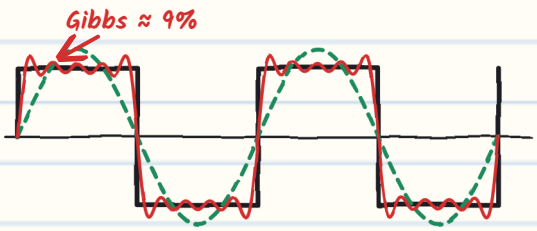
5 Worked Example: the Square Wave

written favourite

Odd square wave, $x(t) = +A$ for $0 < t < T_0/2$ and $-A$ for $T_0/2 < t < T_0$.

1. Odd ⇒ $a_0 = a_n = 0$; half-wave symmetric ⇒ only odd n survive
2. $b_n = (2/T_0)[\int_0^{T0/2} A \sin n\omega_0 t\,dt - \int_{T0/2}^{T0} A \sin n\omega_0 t\,dt] = (2A/n\pi)(1 - \cos n\pi)$
3. n odd: $b_n =$ **$4A/(n\pi)$**; n even: $b_n = 0$

Result $x(t) = (4A/\pi)\,[\sin \omega_0 t + (1/3)\sin 3\omega_0 t + (1/5)\sin 5\omega_0 t + \ldots]$



n = 1…9 sum fundamental $(4A/\pi)$

- Total power $P = A^2$ ($|x| = A$ always)
- Fundamental: $(4A/\pi)^2/2 = 8A^2/\pi^2 =$ **81.1%** of P
- n = 1, 3, 5 together: **93.3%**
- Amplitudes fall as **1/n** (a jump); a triangle wave (a kink) as **$1/n^2$** $(8A/(\pi^2 n^2))$; FT: rect → sinc ∝ 1/f, triangle → $\text{sinc}^2 \propto 1/f^2$. Smoother ⇒ narrower spectrum

Gibbs phenomenon Near a jump a truncated series overshoots by **≈ 9% of the jump** (peak ≈ 1.18A). More terms make the ripple narrower, **never smaller**.

6 Rectangular Pulse Train & Power from the Series

Pulses of height A, width τ, period T_0; duty cycle $d = \tau/T_0$:

$$c_n = A \cdot d \cdot \mathrm{sinc}(n \cdot d)$$

- lines every $f_0 = 1/T_0$; DC $= A \cdot d$
- sinc envelope with **nulls at multiples of $1/\tau$**
- $d = 1/4 \rightarrow$ lines $n = \pm 4, \pm 8 \ldots$ are missing

Parseval for the series

$P = (1/T_0)\int_{T_0} |x|^2\, dt = \sum |c_n|^2$

$= a_0^2 + \sum (a_n^2 + b_n^2)/2 = C_0^2 + \sum C_n^2/2$

each harmonic's power adds independently (orthogonality)

Worked example ✍️ $x(t) = 1 + 2\cos(2\pi \cdot 50t) + 4\sin(2\pi \cdot 150t)$: $T_0 = 20$ ms; $a_0 = 1$, $a_1 = 2$, $b_3 = 4 \rightarrow c_{\pm 1} = 1$, $c_{\pm 3} = \mp 2j$ ($|c_3| = 2$); $P = 1 + 2^2/2 + 4^2/2 =$ **11 W**.

Exam trap Keep τ fixed and let $T_0 \rightarrow \infty$: lines crowd together (spacing $1/T_0 \rightarrow 0$) and the line spectrum becomes the **continuous** Fourier transform. That is exactly how the FT is born (below).

7 The Fourier Transform

high-yield

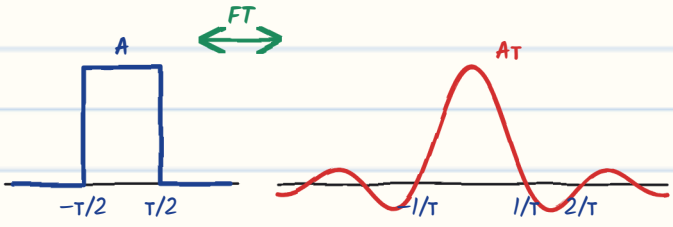
Transform pair $X(f) = \int x(t)\, e^{-j2\pi ft}\, dt$ · $x(t) = \int X(f)\, e^{j2\pi ft}\, df$ (limits $-\infty$ to ∞)

In ω: $X(\omega) = \int x\, e^{-j\omega t}\, dt$, $x(t) = (1/2\pi) \int X(\omega)\, e^{j\omega t}\, d\omega$

- For **aperiodic** signals; $X(f)$ is a continuous spectral density (V/Hz): $|X(f)|$ = amplitude spectrum, $\angle X(f)$ = phase spectrum
- Real $x(t) \Rightarrow X(-f) = X^*(f)$: **$|X|$ is even, phase is odd**
- Exists if $\int |x|\, dt < \infty$ (+ Dirichlet); every energy signal has one. Power signals (DC, cos, $u(t)$) need impulses $\delta(f)$

Sinc & the "sampling function"

$\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$ (normalised) · $\mathrm{Sa}(x) = \sin x / x$, the **sampling function** of the syllabus. $\mathrm{sinc}(0) = \mathrm{Sa}(0) = 1$; sinc zeros at $x = \pm 1, \pm 2 \ldots$; Sa zeros at $\pm\pi, \pm 2\pi \ldots$; $\mathrm{sinc}(x) = \mathrm{Sa}(\pi x)$; $\int \mathrm{sinc}(x)\, dx = 1$.



$A\,\mathrm{rect}(t/\tau) \leftrightarrow A\tau\,\mathrm{sinc}(f\tau)$: nulls every $1/\tau$

→ Key transform pairs (memorise the starred ones)

x(t)	X(f)
$\delta(t)$ ★	1
1 (DC) ★	$\delta(f)$
$\delta(t - t_0)$	$e^{-j2\pi f t_0}$
$e^{j2\pi f_0 t}$	$\delta(f - f_0)$
$\cos 2\pi f_0 t$ ★	$\tfrac{1}{2}[\delta(f - f_0) + \delta(f + f_0)]$
$\sin 2\pi f_0 t$	$(1/2j)[\delta(f - f_0) - \delta(f + f_0)]$
$u(t)$	$1/(j2\pi f) + \tfrac{1}{2}\delta(f)$
$\mathrm{sgn}(t)$	$1/(j\pi f)$

x(t)	X(f)
$A\ \mathrm{rect}(t/\tau)$ ★	$A\tau\ \mathrm{sinc}(f\tau)$
$2W\ \mathrm{sinc}(2Wt)$ ★	$\mathrm{rect}(f/2W)$ (band $\pm W$)
$\Lambda(t/\tau)$ triangle	$\tau\ \mathrm{sinc}^2(f\tau)$
$e^{-at}u(t)$, $a > 0$ ★	$1/(a + j2\pi f)$
$e^{-a\lvert t\rvert}$	$2a/(a^2 + 4\pi^2 f^2)$
$e^{-\pi t^2}$ (Gaussian)	$e^{-\pi f^2}$ (itself)
$\Sigma\ \delta(t - nT_0)$	$f_0\ \Sigma\ \delta(f - nf_0)$

$\Lambda(t/\tau)$: triangle of height 1 and half-width τ (base 2τ). $\mathrm{rect}(t/\tau)$: height 1 for $|t| < \tau/2$.

Exam trap $\delta(t)$ (shortest signal) has a **flat spectrum containing every frequency**; DC (longest) has only $f = 0$. Books use both $\mathrm{sinc}(x) = \sin \pi x/\pi x$ and $\sin x/x$: check which one the options use.

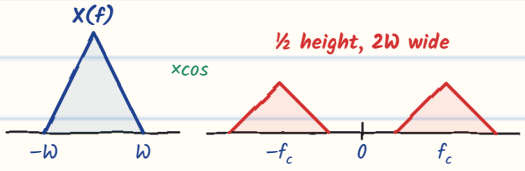
8 Properties of the Fourier Transform

high-yield

Property	x(t)	X(f)
Linearity	$a\,x_1(t) + b\,x_2(t)$	$a\,X_1(f) + b\,X_2(f)$
Time shift ★	$x(t - t_0)$	$X(f)\,e^{-j2\pi f t_0}$ ($\lvert X\rvert$ unchanged)
Frequency shift ★	$x(t)\,e^{j2\pi f_0 t}$	$X(f - f_0)$
Modulation ★	$x(t)\cos 2\pi f_c t$	$\tfrac{1}{2}[X(f - f_c) + X(f + f_c)]$
Time scaling ★	$x(at)$	$(1/\lvert a\rvert)\,X(f/a)$
Time reversal	$x(-t)$	$X(-f)$
Duality ★	$X(t)$	$x(-f)$
Conjugation	$x^*(t)$	$X^*(-f)$
Differentiation	dx/dt	$j2\pi f\,X(f)$
Integration	$\int_{-\infty}^{t} x(\lambda)\,d\lambda$	$X(f)/(j2\pi f) + \tfrac{1}{2}X(0)\delta(f)$
Convolution ★	$x_1(t) * x_2(t)$	$X_1(f) \cdot X_2(f)$
Multiplication ★	$x_1(t) \cdot x_2(t)$	$X_1(f) * X_2(f)$
Area / value at 0	$\int x(t)\,dt = X(0)$	$x(0) = \int X(f)\,df$

Memory tricks ① "A **delay only twists the phase**; a frequency shift is modulation." ② "**Squeeze time, stretch spectrum**": $x(2t)$ has twice the bandwidth. ③ Duality: rect ↔ sinc works both ways.

Worked examples ① Pulse from 0 to T = $A\,\mathrm{rect}((t - T/2)/T) \leftrightarrow AT\,\mathrm{sinc}(fT)\,e^{-j\pi fT}$ (same $|X|$, linear phase). ② RF pulse $A\,\mathrm{rect}(t/T)\cos 2\pi f_c t \leftrightarrow (AT/2)[\mathrm{sinc}((f - f_c)T) + \mathrm{sinc}((f + f_c)T)]$. ③ If $x(t) \leftrightarrow X(f)$, then $x(3t) \leftrightarrow (1/3)X(f/3)$: bandwidth × 3. AM/DSB-SC use ②: see the Modulation lesson.



modulation: the baseband spectrum moves to $\pm f_c$

9 Bandwidth

- **Absolute:** $X(f) = 0$ outside B (ideal band-limited signals only)
- **3-dB (half-power):** where $|X|$ falls to $1/\sqrt{2}$ of peak
- **Null-to-null / first-null:** width of the main lobe
- **99% (fractional-power):** band holding 99% of the power

Signal / system	Bandwidth
pulse, width T	first null $1/T$
NRZ data, R bit/s	first null R Hz
RZ (50% duty) data	first null $2R$ Hz
sinc pulses, R symbol/s	$R/2$ (minimum)
RC low-pass filter	$f_{3dB} = 1/(2\pi RC)$
$x(t)\cos 2\pi f_c t$, x band W	$2W$ around f_c

Time-bandwidth product $T \cdot B \approx$ constant. A pulse of width T has first-null bandwidth $1/T$ (baseband): 1 µs → 1 MHz, 1 ns → 1 GHz. No signal is both time-limited and band-limited.

Exam trap Baseband bandwidth counts **positive** frequencies only: nulls at $\pm 1/T$ give $B = 1/T$, not $2/T$. After modulation the band becomes $2/T$ around f_c.

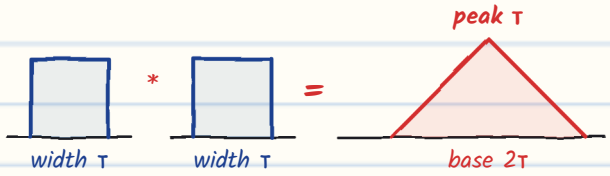
RC: $R = 1\ \mathrm{k\Omega}$, $C = 0.1\ \mu\mathrm{F} \rightarrow f_{3dB} \approx 1.59$ kHz.

10 Convolution

written favourite

Definition $y(t) = x(t) * h(t) = \int x(\tau)\,h(t - \tau)\,d\tau$. For a **linear time-invariant (LTI)** system with impulse response $h(t)$, the output is the input convolved with $h(t)$. Discrete: $y[n] = \sum_k x[k]\,h[n - k]$.

1. **Flip** $h(\tau) \rightarrow h(-\tau)$
2. **Shift** by $t \rightarrow h(t - \tau)$
3. **Multiply** by $x(\tau)$
4. **Integrate** (area) → $y(t)$; repeat for every t



rect * rect = triangle (widths add)

- Commutative, associative, distributive
- $x * \delta(t) = x(t)$ (identity) · $x * \delta(t - t_0) = x(t - t_0)$
- Width of y = width of x + width of h
- Area of y = area of x × area of h
- Discrete length: **$N + M - 1$**

Convolution theorem

$x * h \leftrightarrow X \cdot H$ and $x \cdot h \leftrightarrow X * H$ ⇒ $H(f) = Y(f)/X(f)$ = **transfer function** = FT of $h(t)$. Filters **multiply** spectra; sampling and windowing (products in time) **convolve** spectra.

Worked example 1 ✍ $[1, 2, 3] * [1, 1]$: $y[0] = 1 \cdot y[1] = 2 + 1 = 3 \cdot y[2] = 3 + 2 = 5 \cdot y[3] = 3$

$y =$ **$[1, 3, 5, 3]$**, length $3 + 2 - 1 = 4$ ✓

Worked example 2 ✍ $u(t) * e^{-at}u(t) = \int_0^t e^{-a\tau}\,d\tau =$ **$(1/a)(1 - e^{-at})\,u(t)$**: the step response of an RC low-pass ($a = 1/RC$).

Exam trap Squaring a signal doubles its bandwidth: $x^2 \leftrightarrow X * X$, so a 3 kHz signal squared occupies 6 kHz. A product x_1x_2 occupies $W_1 + W_2$.

→ Correlation: convolution without the flip

Definition $R_{xy}(\tau) = \int x(t)\,y^*(t - \tau)\,dt = x(\tau) * y^*(-\tau)$: **no flip, not commutative** ($R_{xy}(\tau) = R_{yx}^*(-\tau)$), $F\{R_{xy}\} = X \cdot Y^*$. **Autocorrelation** $R_x(\tau)$: even for real x, maximum at $\tau = 0$, $R(0) = \text{energy}$, $F\{R_x\} = |X(f)|^2$. IT link: radar, GPS and CDMA receivers find a delay at the **correlation peak** (matched filter).

11 Parseval's Theorem *high-yield*

Statement The energy (or power) of a signal is the same whether summed over time or over frequency. Each frequency component contributes independently, because the complex exponentials are orthogonal.

Case	Time domain	Frequency domain
FT, energy (Rayleigh) ★	$E = \int \lvert x(t)\rvert^2\,dt$	**$= \int \lvert X(f)\rvert^2\,df$**
Fourier series, power ★	$P = (1/T_0)\int_{T_0} \lvert x(t)\rvert^2\,dt$	$= \sum \lvert c_n\rvert^2 = C_0^2 + \sum C_n^2/2$
DFT	$\sum \lvert x[n]\rvert^2$	$= (1/N)\sum \lvert X[k]\rvert^2$

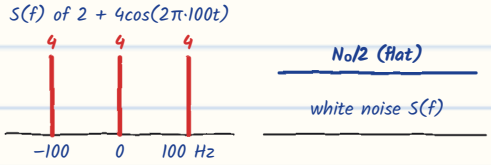
Worked examples ✍

1. $x(t) = 2 + 4\cos(2\pi \cdot 100t)$: $c_0 = 2$, $c_{\pm 1} = 2 \rightarrow P = 2^2 + 2^2 + 2^2 =$ **12 W** $(= 2^2 + 4^2/2$ ✓)
2. $\int \text{sinc}^2(t)\,dt = \int \text{rect}^2(f)\,df =$ **1**: integrate in whichever domain is easier
3. $e^{-at}u(t)$: $E = \int_0^\infty e^{-2at}\,dt = 1/(2a) = \int df/(a^2 + 4\pi^2f^2)$ ✓; $a = 2 \rightarrow$ **0.25 J**. A rect pulse keeps **≈ 90%** of its energy in the main lobe $|f| < 1/\tau$

12 Energy & Power Spectral Density

MCQ + Written

	ESD (energy signals)	PSD (power signals)
Definition	$\Psi(f) = \lvert X(f)\rvert^2$ (J/Hz)	$S(f) = \lim_{T\to\infty} \lvert X_T(f)\rvert^2/T$ (W/Hz)
Autocorrelation	$R(\tau) = \int x(t)\, x^*(t-\tau)\, dt$; $R(0) = E$	time-averaged $R(\tau)$; **$R(0) = P$**
Wiener–Khinchin	$\Psi(f) = \mathcal{F}\{R(\tau)\}$	$S(f) = \mathcal{F}\{R(\tau)\}$
Through $H(f)$	$\Psi_y = \lvert H(f)\rvert^2 \Psi_x$	$S_y = \lvert H(f)\rvert^2 S_x$



periodic signal = PSD lines; white noise = flat PSD

Periodic: $S(f) = \sum |c_n|^2\, \delta(f - nf_0)$ (lines). **White noise:** $S(f) = N_0/2$, $R(\tau) = (N_0/2)\delta(\tau)$; after an ideal LPF of bandwidth B the noise power is N_0B (the N in Shannon's $C = B\log_2(1 + S/N)$).

Exam trap $|X(f)|^2$ is the **energy** spectral density; for a power signal divide by T and take the limit. PSD is real, even, ≥ 0: **phase is lost**. Remember: "**R and S are a Fourier pair; R(0) is the total.**"

Worked examples

1. $A\cos(2\pi f_0 t + \Theta)$: $R(\tau) = (A^2/2)\cos 2\pi f_0\tau \to S(f) = (A^2/4)[\delta(f - f_0) + \delta(f + f_0)]$; $P = R(0) = \mathbf{A^2/2}$ (Θ is lost)
2. White noise $N_0/2$ through an RC low-pass: $S_y = (N_0/2)/(1 + (2\pi fRC)^2) \to P_y = N_0/(4RC)$; noise-equivalent bandwidth $B_N = 1/(4RC) = (\pi/2)\, f_{3dB}$

13 The Sampling Theorem

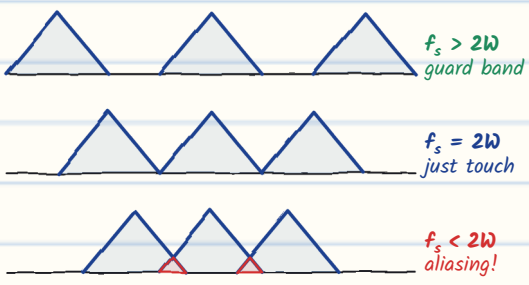
high-yield

Statement A signal **band-limited to W Hz** ($X(f) = 0$ for $|f| > W$) is fully recovered (ideal LPF) from samples taken at $f_s \geq 2W$; Nyquist rate $= 2W$ (minimum f_s); Nyquist interval $= 1/(2W)$ (maximum T_s). Nyquist 1928, Shannon 1949.

Ideal (impulse) sampling: $x_s(t) = x(t) \cdot \sum \delta(t - nT_s)$

$\Rightarrow$ $X_s(f) = f_s \sum X(f - n f_s)$: copies of the spectrum at every multiple of f_s (a product in time is a convolution in frequency).

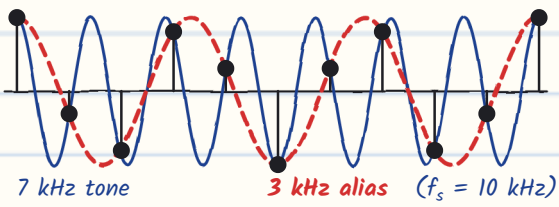
Recovery: LPF cut-off $W \leq f_c \leq f_s - W$; guard band $= f_s - 2W$. Ideal interpolation: $x(t) = \sum x(nT_s)\, \mathrm{sinc}((t - nT_s)/T_s)$.



spectral copies every f_s: gap, touch, overlap

Aliasing: if $f_s < 2W$ the copies overlap; a tone f appears at $f_a = |f - k \cdot f_s|$ (k = nearest integer, $f_a \le f_s/2$). Cure: **anti-aliasing LPF** (cut-off $f_s/2$) **before** the sampler + sample above $2W$.

Practical: natural (gated) sampling keeps the baseband copy undistorted; **flat-top** (sample-and-hold, PAM) multiplies it by $\text{sinc}(f\tau)$ → aperture effect, fixed by an equaliser.



same samples, two sinusoids: the LPF rebuilds 3 kHz

➜ Nyquist-rate & alias problems (high-yield MCQs)

Signal	Working	Answer
band-limited to 4 kHz	2 × 4 kHz	**8 kHz**
$\cos(200\pi t) + \cos(600\pi t)$	100 Hz, 300 Hz → f_{max} = 300 Hz	**600 Hz**
speech, 3.4 kHz	T_s = 1/6800 ≈ 147 µs; telephony samples at 8 kHz	**6.8 kHz**
CD audio, 20 kHz	uses 44.1 kHz (guard band 4.1 kHz)	**40 kHz**
$\cos^2(2\pi \cdot 1000t)$	= ½ + ½cos(2π·2000t) → f_{max} = 2 kHz	**4 kHz**
x_1(3 kHz) · x_2(5 kHz)	product: W = 3 + 5 = 8 kHz (a sum: 5 kHz)	**16 kHz**
sinc(200t); $\text{sinc}^2(200t)$	W = 100 Hz; squaring doubles W to 200 Hz	**200; 400 Hz**
60 Hz tone, f_s = 100 Hz	|60 − 100| = 40 Hz	alias **40 Hz**

Exam trap A pure tone sampled at **exactly** $2f$ can give all-zero samples ($\sin n\pi = 0$), so use $f_s > 2f$ strictly. Chain: anti-aliasing LPF → sampler → quantizer → encoder (PCM details: Modulation lesson).

14 DFT & FFT (the digital spectrum)

DFT $X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}$, $k = 0 \dots N-1$ · inverse has the factor **1/N** · bin $k \leftrightarrow f = k \cdot f_s/N$ · resolution $\Delta f = f_s/N$

- Radix-2 FFT (Cooley–Tukey, **1965**, N a power of 2): $(N/2)\log_2 N$ complex multiplications instead of N^2
- N = 1024: 1,048,576 → 5,120, ≈ **205x faster**
- f_s = 8 kHz, N = 256 → Δf = **31.25 Hz**
- DFT product ↔ **circular** convolution
- IT link: OFDM (Wi-Fi, 4G/5G) uses IFFT/FFT

Worked example ✍ 4-point DFT of x = [1, 1, 0, 0]: $X[k] = 1 + e^{-j\pi k/2}$ → X = [2, 1 − j, 0, 1 + j]. Parseval check: $\Sigma|x|^2 = 2 = (1/4)(4 + 2 + 0 + 2)$ ✓. $X[0] = \Sigma\, x[n]$ is always the DC sum.

Night-Before-the-Exam Sheet

Formula sheet

$E = \int |x|^2\,dt$ · $P = \lim (1/T)\int |x|^2\,dt$

$f_0 = 1/T_0$ · $\omega_0 = 2\pi/T_0$

$c_n = (a_n - jb_n)/2$ · $|c_n| = C_n/2$

$X(f) = \int x\, e^{-j2\pi ft}\,dt$

$A\,\mathrm{rect}(t/\tau) \leftrightarrow A\tau\,\mathrm{sinc}(f\tau)$

$x(t - t_0) \leftrightarrow X\, e^{-j2\pi f t_0}$

$x \cos 2\pi f_c t \leftrightarrow \tfrac{1}{2}[X(f-f_c) + X(f+f_c)]$

$E = \int |x|^2\,dt = \int |X|^2\,df$

$f_s \geq 2W$ · $T_s \leq 1/(2W)$

$A\cos(\omega_0 t + \theta) \rightarrow P = A^2/2$ · DC $A \rightarrow A^2$

$a_0 = (1/T_0)\int x$ · a_n, b_n: $(2/T_0)\int \ldots$

Square wave: $b_n = 4A/(n\pi)$, odd n

$x(t) = \int X\, e^{j2\pi ft}\,df$

$\delta(t) \leftrightarrow 1$ · $1 \leftrightarrow \delta(f)$

$x(at) \leftrightarrow (1/|a|)X(f/a)$

$x * h \leftrightarrow XH$ · $xh \leftrightarrow X * H$

$P = \Sigma |c_n|^2$ · $S(f) = \mathcal{F}\{R(\tau)\}$

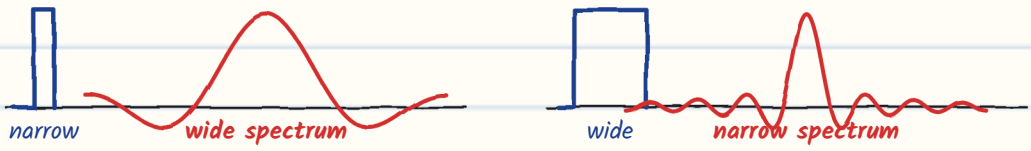
$f_a = |f - k f_s|$ · $\Delta f = f_s/N$

→ Top 12 one-liners for MCQs

1. Even signal → only DC + cosines
2. Odd signal → only sines
3. Half-wave symmetry → odd harmonics
4. Square wave harmonics ∝ 1/n
5. Gibbs overshoot ≈ 9%, never vanishes
6. FT of rect = sinc; of $\delta(t) = 1$
7. Convolution in t = multiplication in f
8. Parseval: energy same in t and f
9. PSD = FT of autocorrelation (Wiener–Khinchin)
10. Nyquist rate = $2f_{max}$; 4 kHz → 8 kHz
11. Anti-aliasing filter goes before the sampler
12. Periodic signal = power signal

Top traps two-sided lines are half height · $|X(f)|^2$ is ESD, not PSD · $\cos^2$ and products widen bandwidth · Nyquist **rate** $(2W)$ ≠ Nyquist **interval** $(1/2W)$ · tone at exactly 2f: use $f_s > 2f$ · zero-padding does not improve resolution · two sinc conventions

All tricks together Energy signals die out, power signals live for ever · Even ↔ cos, Odd ↔ sin · half-wave keeps odd harmonics · a delay only twists the phase · squeeze time, stretch spectrum · R and S are a Fourier pair, R(0) is the total



time-bandwidth product: $T \cdot B \approx$ constant

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. Nyquist rate of $x(t) = 5\cos(1000\pi t) + 3\sin(3000\pi t)$:

(a) 1500 Hz (b) 3000 Hz
(c) 1000 Hz (d) 6000 Hz

2. The Fourier transform of $\delta(t)$ is:

(a) $\delta(f)$ (b) 0
(c) 1 (d) sinc(f)

3. The Fourier series of an odd periodic function has:

(a) only cosine terms (b) only a DC term
(c) DC and cosines (d) only sine terms

4. Convolution in time corresponds in frequency to:

(a) multiplication (b) convolution
(c) addition (d) differentiation

5. Average power of $3\cos(100\pi t) + 4\sin(200\pi t)$:

(a) 25 W (b) 12.5 W
(c) 7 W (d) 3.5 W

6. If $x(t) \leftrightarrow X(f)$, then $x(t - t_0) \leftrightarrow$

(a) $X(f - t_0)$ (b) $X(f)\, e^{j2\pi f t_0}$
(c) $X(f)\, e^{-j2\pi f t_0}$ (d) $X(f)/t_0$

7. A symmetrical square wave contains:

(a) all harmonics (b) only even harmonics
(c) odd harmonics $\propto 1/n^2$ (d) odd harmonics $\propto 1/n$

8. A 60 Hz tone sampled at 100 Hz is reconstructed as:

(a) 40 Hz (b) 60 Hz
(c) 160 Hz (d) 100 Hz

9. Energy of the pulse $A\,\mathrm{rect}(t/\tau)$:

(a) $A\tau$ (b) $A^2\tau$
(c) $A^2/2$ (d) $A^2\tau^2$

10. By Wiener–Khinchin, the PSD is the FT of the:

(a) signal itself (b) impulse response
(c) phase spectrum (d) autocorrelation

Answer key → 1-b · 2-c · 3-d · 4-a · 5-b · 6-c · 7-d · 8-a · 9-b · 10-d

Written question Find the Fourier transform of the pulse $x(t) = A$ for $|t| < \tau/2$ (0 elsewhere) and sketch its amplitude spectrum. Using Parseval's theorem, find its energy and first-null bandwidth for $A = 2$ V, $\tau = 1$ ms. 8 marks

1. $X(f) = \int_{-\tau/2}^{\tau/2} A\, e^{-j2\pi f t}\, dt = A\cdot[e^{j\pi f\tau} - e^{-j\pi f\tau}]/(j2\pi f) = A\,\sin(\pi f\tau)/(\pi f) = \mathbf{A\tau\ sinc(f\tau)}$ — 3
2. Sketch: real, even; peak $A\tau$ at $f = 0$, nulls at $\pm 1/\tau$, $\pm 2/\tau$ …, side lobes fall as $1/f$ — 1
3. Parseval: $E = A^2\tau = \int A^2\tau^2\,\mathrm{sinc}^2(f\tau)\,df$ ($\int \mathrm{sinc}^2(f\tau)\,df = 1/\tau$) — 2; $A = 2$ V, $\tau = 1$ ms: $E =$ **4 mJ**, first null $1/\tau =$ **1 kHz** ($\approx 90\%$ of E inside) — 2

→ Before the exam ✓

- ✓ Can classify energy / power signals
- ✓ Can write a_0, a_n, b_n, c_n and use symmetry
- ✓ Know 8 FT pairs and the shift, scaling, modulation rules
- ✓ Can convolve two short sequences
- ✓ Can find a Nyquist rate and an alias frequency
- ✓ Also practise: triangle-wave series, proof of Parseval, rect * rect